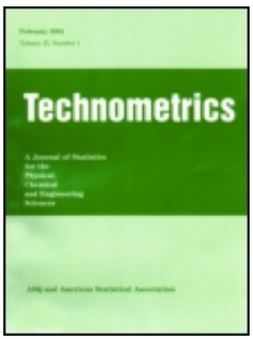

Technometrics

ISSN: 0040-1706 (Print) 1537-2723 (Online) Journal homepage: http://www.tandfonline.com/loi/utch20

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To cite this article: James W. Frane (1977) A Note on Checking Tolerance in Matrix Inversion and Regression, Technometrics, 19:4, 513-514

To link to this article: http://dx.doi.org/10.1080/00401706.1977.10489595

Published online: 09 Apr 2012.

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TECHNOMETRICS©, VOL. 19, NO. 4, NOVEMBER 1977

A Note on Checking Tolerance in Matrix Inversion and Regression

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Efroymsen's method of checking tolerance in matrix inversion is a popular technique to avoid inverting singular and nearly singular covariance matrices. Recently Berk showed that the Efroymsen tolerance check is theoretically unsatisfactory. A simple modification of the Efroymsen tolerance check is given here that provides a proper tolerance test.

KEY WORDS

Inversion
pivot
regression
singular
sweep
tolerance

Efroymsen [2], provides a method of checking for singularity and near singularity in a covariance matrix as part of the process of doing stepwise regression. Essentially, the Efroymsen tolerance check computes a regression by including predictor variables one at a time in such a way that no predictor is included if its squared multiple correlation with previously included variables exceeds one minus a specified tolerance limit. Thus, the tolerance limit is the minimum permissible proportion of the variance in a predictor that remains after regressing it on already included predictors.

The problem with this tolerance check is that no check is made on the effect a newly included variable has on those already included. It was hoped that, in practice, this effect would not be large. Berk [1], has shown that this effect can be made arbitrarily large (i.e., the squared multiple correlation of a predictor variable with the other predictor variables can be made arbitrarily close to one without violating the Efroymsen tolerance test), and that the regression coefficient can be totally inaccurate.

A number of methods are available to detect this unhappy event. You can compute the condition number of the correlation matrix of the predictor variables as a better test of singularity, but this is not an economical procedure in the midst of stepwise regression. You could also compute the final tolerance of each predictor variable, i.e., the squared multiple correlation of each predictor with all other predictors after all predictors have been selected. This does not provide a good regression equation, but it does identify problem variables. Or you might consider removing predictors that have low final tolerances, but the damage has already been done—you cannot recover by performing reverse sweeps or pivots.

Problem variables must be identified before they are included. Suppose, without loss of generality, that we have included variables $X_2, \cdots, X_p$, and are considering the inclusion of X_1. The Efroymsen tolerance test considers the squared multiple correlation of X_1 with $X_2, \cdots, X_p$. A proper tolerance test also considers the squared multiple correlation of each X_i, $i = 1, \cdots, p$ with $X_1, \cdots, X_{i-1}, \cdots, X_p$. X_1 should not be included if any of these squared multiple correlations exceed one minus the tolerance limit.

The amount of added computation is small.

Let

$$S = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix}$$

be the sample covariance matrix where S_{11} is the submatrix for X_1 and S_{22} is the submatrix for $X_2, \cdots, X_p$. (A parallel discussion could be given for a correlation matrix or matrix of cross products of deviations.) Let C be the covariance matrix for the predictor variables after $X_2, \cdots, X_p$ have been included; i.e., after sweeping has been performed for these variables.

Then

$$C = \begin{pmatrix} C_{11} & C_{21} \\ C_{21} & C_{22} \end{pmatrix} = \begin{pmatrix} S_{11} - S_{12}S_{22}^{-1}S_{21} & S_{12}\ S_{22}^{-1} \\ S_{22}^{-1}S_{21} & -S_{22}^{-1} \end{pmatrix}$$

Received February 1977; revised April 1977

(A negative sign is used in front of S_{22}^{-1} because the sweep operator is used rather than the pivot.) If we sweep for X_1, the new values of the diagonal corresponding to $X_2, \cdots, X_p$ are obtained as the diagonal elements of

$$C_{22} - C_{21}C_{11}^{-1}C_{12}$$

Let these diagonal elements be denoted by $d_2, \cdots, d_p$. The proper tolerance test prohibits inclusion of X_i whenever $(-1/s_{ii}d_i)$ is less than the tolerance limit for any i, $i = 2, \cdots, p$. $(-1/s_{ii}d_i)$ is one minus the squared multiple correlation of X_i with all other predictors $X_1, \cdots, X_p$.

The amount of computation for doing a proper tolerance test before sweeping is small: when the number of variables is small, the total amount of computation is small; when the number of variables is large, the percent increase in computation is small. Moreover, since the number of cases usually far exceeds the number of variables, the cost of computing the covariance matrix, residuals and their standard errors, and other statistics that accompany regression will usually far exceed the cost of sweeping. Hence, a proper tolerance test can be made without noticeable increase in cost. Such a tolerance test was incorporated in the August, 1976 release of the BMDP statistical computer programs.

To see how the proper tolerance test works, we consider the first three predictors from the example used by Berk. The correlation matrix for these predictor variables is

$$\begin{bmatrix} 1 & & \\ \sqrt{.99} & 1 & \\ .1\sqrt{.99} & 0 & 1 \end{bmatrix}$$

where the upper triangular portion is omitted because of symmetry and where the correlations of the dependent variable with the predictors can be anything so long as stepwise regression selects the predictors in the order X_1, X_2, X_3. Let the tolerance limit be .001. At step one, there is no tolerance check since there is only one predictor. Sweeping for X_1 gives the matrix

$$\begin{bmatrix} -1 & & \\ \sqrt{.99} & .01 & \\ .1\sqrt{.99} & -.099 & .9901 \end{bmatrix}.$$

At step two, X_2 passes the usual tolerance test because its diagonal element, .01, is greater than .001. Also, using the complete tolerance test, we compute what the diagonal element would be for X_1 if we were to sweep for X_2: $d_1 = -1 - \sqrt{.99} * 100 * \sqrt{.99} = -100$. $(-1/d_1s_{11}) = .01 > .001$ so X_2 passes the complete tolerance test. Sweeping for X_2 gives

$$\begin{bmatrix} -100 & & \\ 10\sqrt{99} & -100 & \\ \sqrt{99} & -9.9 & .01 \end{bmatrix}.$$

X_3 passes the classical tolerance test since its diagonal element is greater than .001. However, we now have $d_1 = -100 - \sqrt{99} * 100 * \sqrt{99} = -10000$ and $(-1/d_1s_{11}) = .0001 < .001$ so X_3 does not pass the complete tolerance test. The squared multiple correlation of X_1 with X_2 and X_3 is .9999.

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